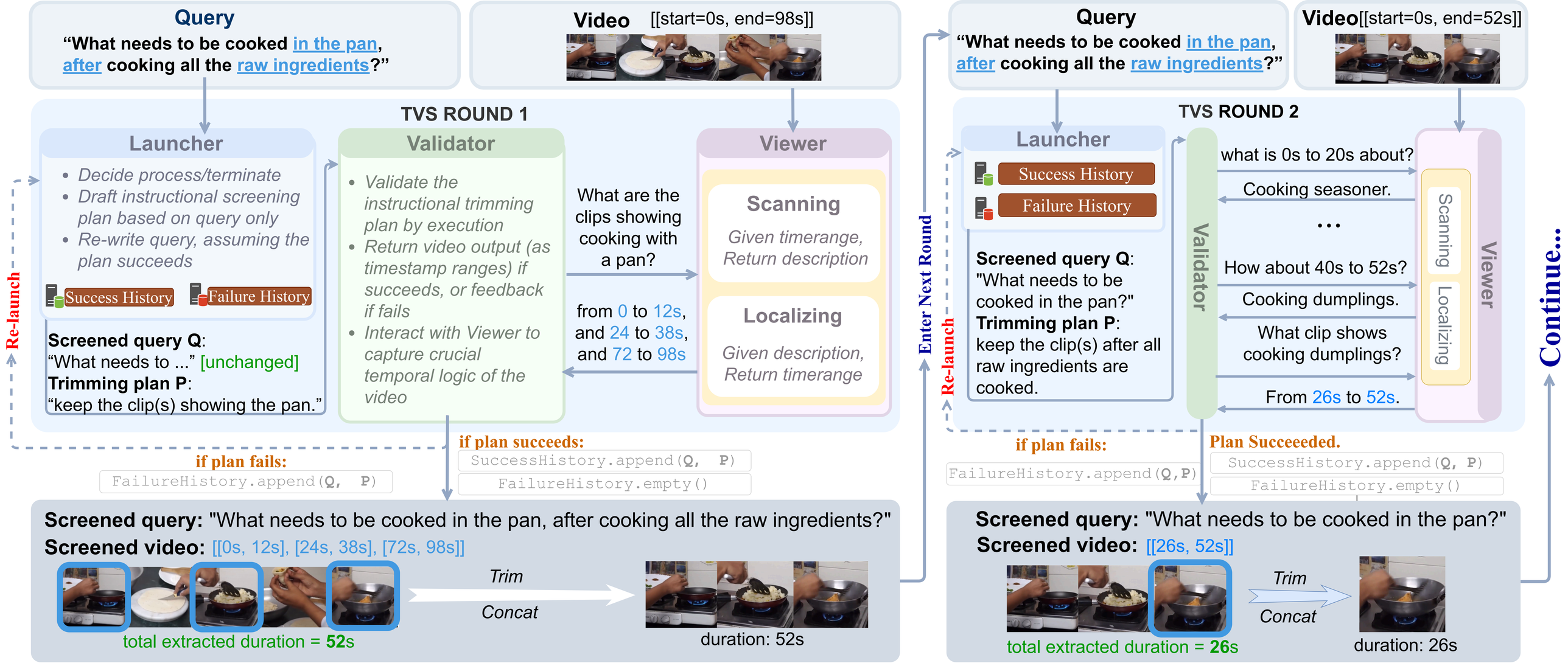
Query
"What needs to be cooked in the pan, after cooking all the raw ingredients?"
Video [[start=0s, end=98s]]
TVS ROUND 1
Launcher
Decide process/terminate
Draft instructional screening plan based on query only
Re-write query, assuming the plan succeeds
Success History
Failure History
Screened query Q: "What needs to ..." [unchanged]
Trimming plan P: "keep the clip(s) showing the pan."
Validator
Validate the instructional trimming plan by execution
Return video output (as timestamp ranges) if succeeds, or feedback if fails
Interact with Viewer to capture crucial temporal logic of the video
What are the clips showing cooking with a pan?
from 0 to 12s, and 24 to 38s, and 72 to 98s
Viewer
Scanning
Given timerange, Return description
Localizing
Given description, Return timerange
Re-launch
if plan fails:
FailureHistory.append(Q, P)
if plan succeeds:
SuccessHistory.append(Q, P)
FailureHistory.empty()
Screened query: "What needs to be cooked in the pan, after cooking all the raw ingredients?"
Screened video: [[0s, 12s], [24s, 38s], [72s, 98s]]
Trim
Concat
total extracted duration = 52s
duration: 52s
Enter Next Round
Query
"What needs to be cooked in the pan, after cooking all the raw ingredients?"
Video[[start=0s, end=52s]]
TVS ROUND 2
Launcher
Success History
Failure History
Screened query Q: "What needs to be cooked in the pan?"
Trimming plan P: keep the clip(s) after all raw ingredients are cooked.
Validator
what is 0s to 20s about?
Cooking seasoner.
...
How about 40s to 52s?
Cooking dumplings.
What clip shows cooking dumplings?
From 26s to 52s.
Scanning
Localizing
Viewer
Re-launch
if plan fails:
FailureHistory.append(Q,P)
Plan Succeeeded.
SuccessHistory.append(Q, P)
FailureHistory.empty()
Screened query: "What needs to be cooked in the pan?"
Screened video: [[26s, 52s]]
Trim
Concat
total extracted duration = 26s
duration: 26s
Continue...